NIS (Network Information Service) Server & Client Configuration

Cluster: Virtual HPC Cluster

NIS Server: hpc-master (192.168.100.1)

NIS Clients: compute1, compute2, login, storage

NIS Domain: hpc.local

1. Info

NIS (Network Information Service) provides **centralized user account management** across all cluster nodes. Instead of maintaining separate user accounts on each node, all user information (passwd, group, shadow) is managed on **hpc-master** and distributed to all client nodes automatically.

```
hpc-master (NIS Server — 192.168.100.1)
          │
          │  NIS Domain: hpc.local
          │  Network: 192.168.100.0/24
          │
    ┌─────┴──────────────────────────┐
    │            │            │       │
compute1  compute2    login  storage
(NIS Client) (NIS Client) (NIS Client) (NIS Client)
```

Services running on NIS Server:

Service	Purpose
ypserv	Main NIS server — serves NIS maps to clients
yppasswdd	Allows users to change passwords via NIS
ypxfrd	Accelerates NIS map transfers to clients
ypbind	Binds the master itself to its own NIS domain

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2. NIS Server Configuration — hpc-master

2.1 Set NIS Domain Name

```
# Set NIS domain name for current session
nisdomainname hpc.local

# Make it persistent across reboots
echo "NISDOMAIN=hpc.local" >> /etc/sysconfig/network

# Verify
nisdomainname
```

Expected output:

```
hpc.local
```

2.2 Install NIS Server Packages

```
dnf install ypserv yp-tools ypbind -y
```

Packages installed:

Package	Purpose
ypserv	NIS server daemon
yp-tools	NIS utilities (ypwhich, ypcat, yppoll)
ypbind	NIS client binder (needed on server too)
nss_nis	NSS plugin for NIS lookups
tokyocabinet	Database backend for NIS maps

2.3 Configure /etc/ypserv.conf

```
vi /etc/ypserv.conf
```

Added at the bottom:

```
# Allow cluster network access
192.168.100.0/24 : * : * : none
```

2.4 Configure /var/yp/securenets

```
vi /var/yp/securenets
```

Contents:

```
# Allow localhost
255.0.0.0       127.0.0.0

# Allow cluster network
255.255.255.0   192.168.100.0
```

Explanation: securenets restricts which hosts are allowed to query the NIS server. Only localhost and the cluster network 192.168.100.0/24 are permitted.

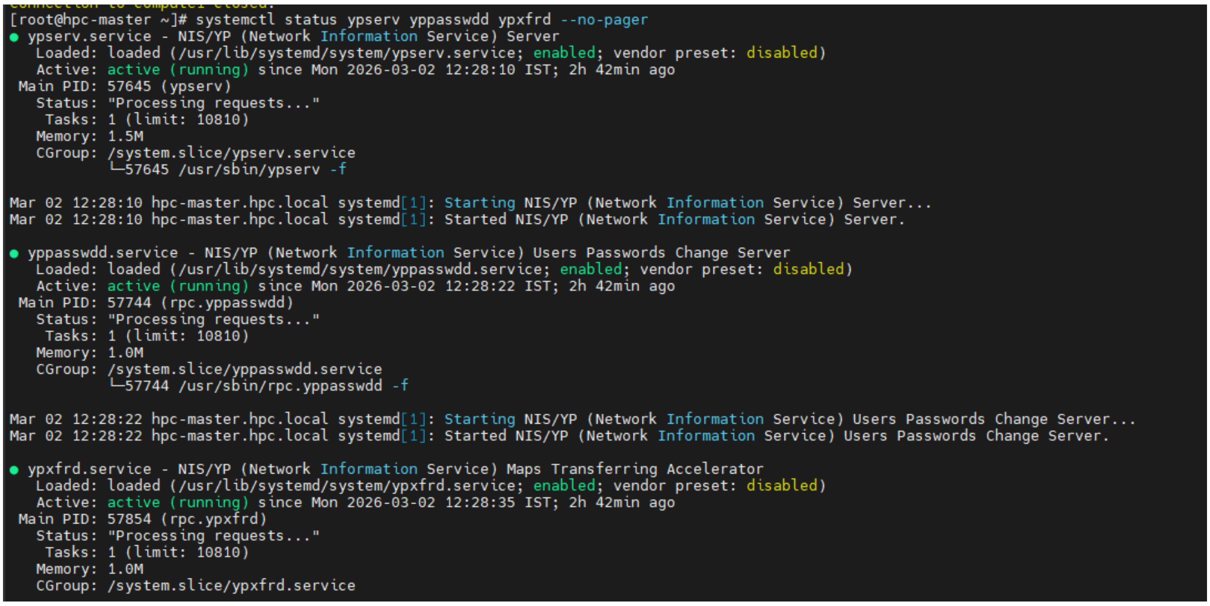
2.5 Start and Enable NIS Services

```
# Enable and start ypserv
systemctl enable --now ypserv

# Enable and start yppasswdd
systemctl enable --now yppasswdd

# Enable and start ypxfrd
systemctl enable --now ypxfrd

# Verify all services
```

```
systemctl status ypserv yppasswdd ypxfrd --no-pager
```

All three services confirmed active and running:

```
ypserv.service    — Active: active (running) — Status: "Processing
requests..."
yppasswdd.service — Active: active (running) — Status: "Processing
requests..."
ypxfrd.service    — Active: active (running) — Status: "Processing
requests..."
```



```
[root@hpc-master ~]# systemctl status ypserv yppasswdd ypxfrd --no-pager
● ypserv.service - NIS/YP (Network Information Service) Server
   Loaded: loaded (/usr/lib/systemd/system/ypserv.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2026-03-02 12:28:10 IST; 2h 42min ago
 Main PID: 57645 (ypserv)
   Status: "Processing requests..."
    Tasks: 1 (limit: 10810)
   Memory: 1.5M
   CGroup: /system.slice/ypserv.service
           └─57645 /usr/sbin/ypserv -f

Mar 02 12:28:10 hpc-master.hpc.local systemd[1]: Starting NIS/YP (Network Information Service) Server...
Mar 02 12:28:10 hpc-master.hpc.local systemd[1]: Started NIS/YP (Network Information Service) Server.

● yppasswdd.service - NIS/YP (Network Information Service) Users Passwords Change Server
   Loaded: loaded (/usr/lib/systemd/system/yppasswdd.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2026-03-02 12:28:22 IST; 2h 42min ago
 Main PID: 57744 (rpc.yppasswdd)
   Status: "Processing requests..."
    Tasks: 1 (limit: 10810)
   Memory: 1.0M
   CGroup: /system.slice/yppasswdd.service
           └─57744 /usr/sbin/rpc.yppasswdd -f

Mar 02 12:28:22 hpc-master.hpc.local systemd[1]: Starting NIS/YP (Network Information Service) Users Passwords Change Server...
Mar 02 12:28:22 hpc-master.hpc.local systemd[1]: Started NIS/YP (Network Information Service) Users Passwords Change Server.

● ypxfrd.service - NIS/YP (Network Information Service) Maps Transferring Accelerator
   Loaded: loaded (/usr/lib/systemd/system/ypxfrd.service; enabled; vendor preset: disabled)
   Active: active (running) since Mon 2026-03-02 12:28:35 IST; 2h 42min ago
 Main PID: 57854 (rpc.ypxfrd)
   Status: "Processing requests..."
    Tasks: 1 (limit: 10810)
   Memory: 1.0M
   CGroup: /system.slice/ypxfrd.service
```

2.6 Initialize NIS Database

```
/usr/lib64/yp/ypinit -m
```

At the prompt, press `Ctrl+D` then confirm with `y`. The following NIS maps were built:

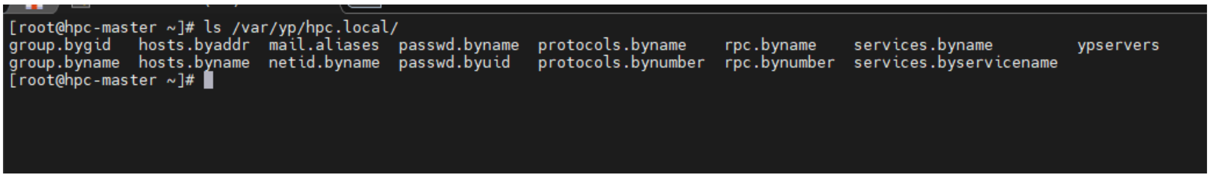
```
Building /var/yp/hpc.local/ypservers...
Updating passwd.byname...
Updating passwd.byuid...
Updating group.byname...
Updating group.bygid...
Updating hosts.byname...
Updating hosts.byaddr...
Updating rpc.byname...
Updating rpc.bynumber...
Updating services.byname...
Updating services.byservicename...
Updating netid.byname...
Updating protocols.bynumber...
Updating protocols.byname...
Updating mail.aliases...
```

Verify maps were created:

```
ls /var/yp/hpc.local/
```

Output:

```
group.bygid   hosts.byaddr  mail.aliases  passwd.byname  protocols.byname
group.byname  hosts.byname  netid.byname  passwd.byuid   protocols.bynumber
rpc.byname    rpc.bynumber  services.byname  services.byservicename
ypservers
```



2.7 Configure ypbind on Master

Since SELinux is disabled on this cluster, the default ypbind service file fails because it tries to run `setsebool allow_ypbind=1`. This was resolved by overriding the service file.

Add server entry to /etc/yp.conf:

```
echo "domain hpc.local server 192.168.100.1" > /etc/yp.conf
```

Override ypbind service to skip setsebool:

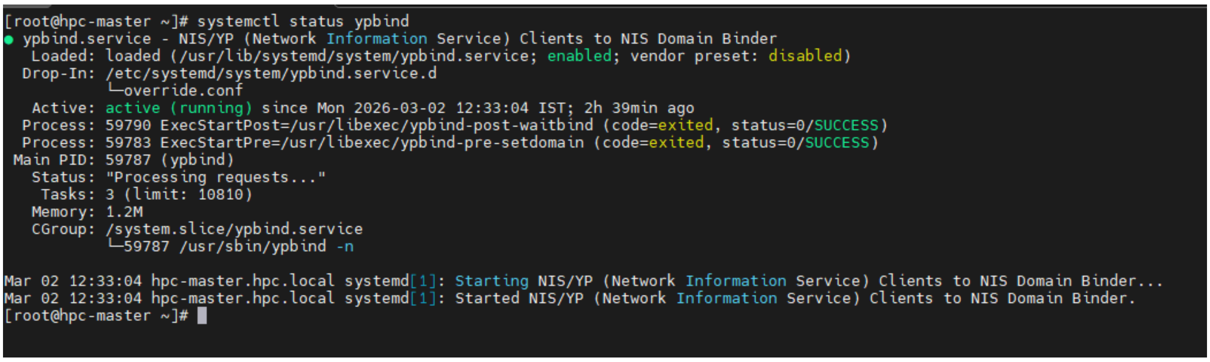
```
mkdir -p /etc/systemd/system/ypbind.service.d/
cat > /etc/systemd/system/ypbind.service.d/override.conf << 'EOF'
[Service]
ExecStartPre=
ExecStartPre=/usr/libexec/ypbind-pre-setdomain
EOF

systemctl daemon-reload
```

Start ypbind:

```
systemctl enable --now ypbind
systemctl status ypbind --no-pager
```

Output:

```
● ypbind.service - NIS/YP (Network Information Service) Clients to NIS
Domain Binder
   Active: active (running) since Mon 2026-03-02 12:33:04 IST
   Status: "Processing requests..."
```



2.8 Verify NIS Server

```
ypwhich
```

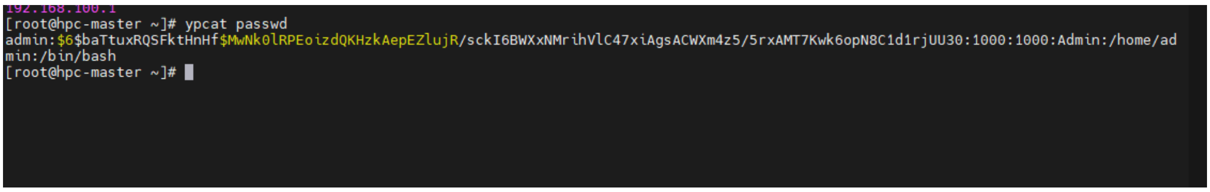
Output:

```
192.168.100.1
```


```
ypcat passwd
```

Output:

```
admin:$6$baTtuxRQ...:1000:1000:Admin:/home/admin:/bin/bash
```



NIS server is fully operational.

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3. NIS Client Configuration

All four client nodes were configured individually via SSH. The same steps were applied to compute1, compute2, login, and storage.

Note: Client nodes were configured by SSHing directly into each node from hpc-master, not via xCAT tools.

3.1 SSH Into Each Client Node

```
# From hpc-master
```

```
ssh root@compute1    # repeat for compute2, login, storage
```

3.2 Install NIS Client Packages

```
dnf install ypbind yp-tools -y
```

3.3 Set NIS Domain Name

```
# Set for current session
nisdomainname hpc.local

# Make persistent across reboots
echo "NISDOMAIN=hpc.local" >> /etc/sysconfig/network
```

3.4 Configure /etc/yp.conf

```
echo "domain hpc.local server 192.168.100.1" > /etc/yp.conf
cat /etc/yp.conf
```

Output:

```
domain hpc.local server 192.168.100.1
```

3.5 Fix ypbind SELinux Issue

Since SELinux is disabled on all nodes, the same override applied on the server was applied on each client:

```
mkdir -p /etc/systemd/system/ypbind.service.d/
cat > /etc/systemd/system/ypbind.service.d/override.conf << 'EOF'
[Service]
ExecStartPre=
ExecStartPre=/usr/libexec/ypbind-pre-setdomain
EOF

systemctl daemon-reload
```

3.6 Configure /etc/nsswitch.conf

This tells the OS to look up users, groups, and passwords from NIS in addition to local files:

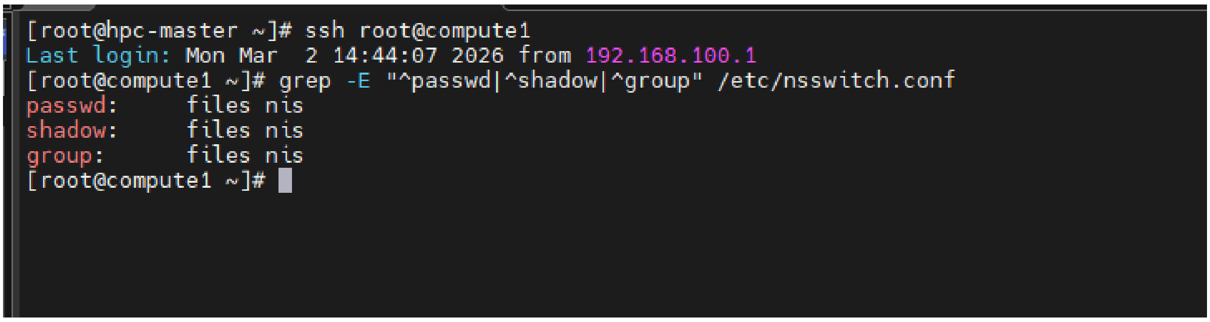
```
sed -i 's/^passwd:.*/passwd:     files nis/' /etc/nsswitch.conf
sed -i 's/^shadow:.*/shadow:     files nis/' /etc/nsswitch.conf
sed -i 's/^group:.*/group:      files nis/' /etc/nsswitch.conf
```

Verify:

```
grep -E "^passwd|^shadow|^group" /etc/nsswitch.conf
```

Expected output:

```
passwd:     files nis
shadow:     files nis
group:      files nis
```



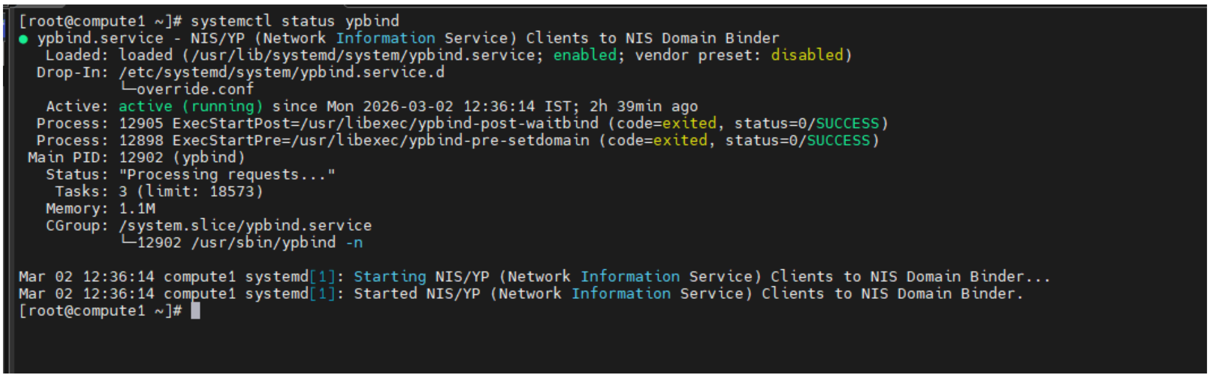
Explanation: `files nis` means the OS checks local `/etc/passwd` first, then falls back to NIS. This ensures local system accounts (root, etc.) always work even if NIS is unavailable.

3.7 Enable and Start ypbind

```
systemctl enable --now ypbind
systemctl status ypbind --no-pager
```

Expected output:

```
● ypbind.service
   Active: active (running)
   Status: "Processing requests..."
```



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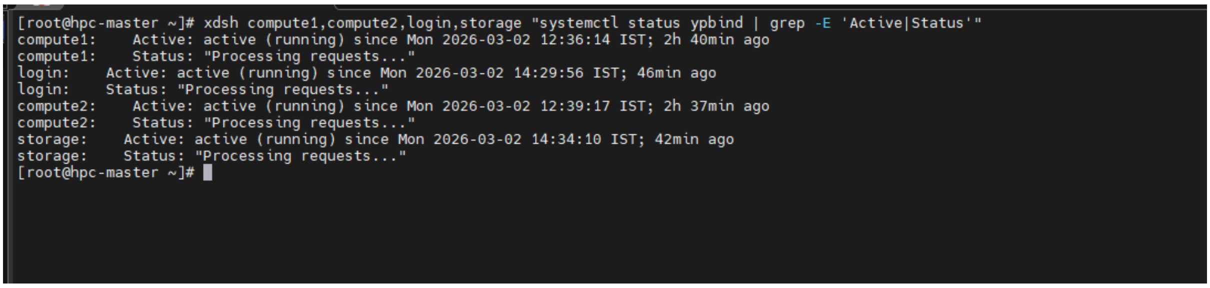
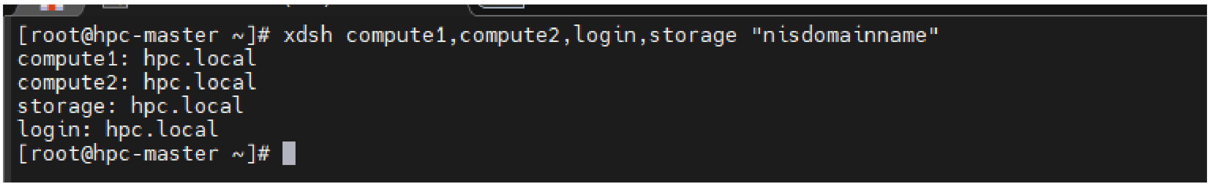
4. Verification

4.1 Verify from hpc-master (All Nodes at Once)

```
# Check ypbind is running on all clients
xdsh compute1,compute2,login,storage "systemctl status ypbind | grep -E
'Active|Status'"
```

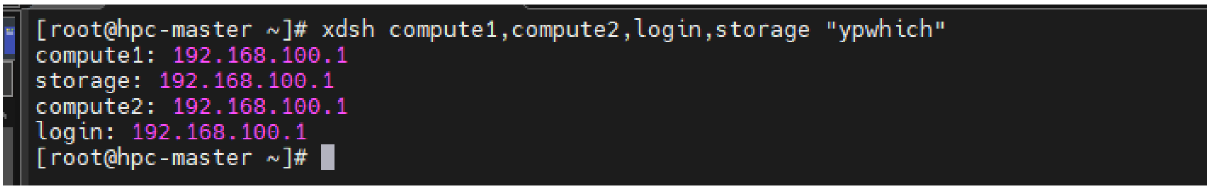


```
# Check NIS domain on all clients
xdsh compute1,compute2,login,storage "nisdomainname"
```



```
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "nisdomainname"
compute1: hpc.local
compute2: hpc.local
storage: hpc.local
login: hpc.local
[root@hpc-master ~]#
```

```
# Check which NIS server each client is bound to
xdsh compute1,compute2,login,storage "ypwhich"
```

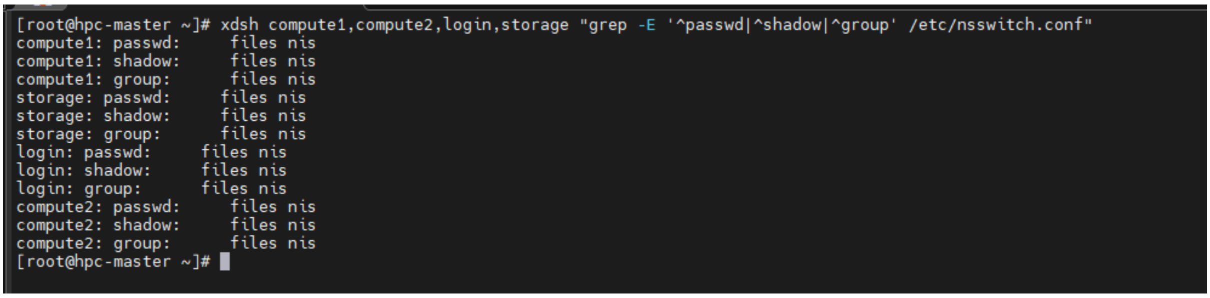


```
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "ypwhich"
compute1: 192.168.100.1
storage: 192.168.100.1
compute2: 192.168.100.1
login: 192.168.100.1
[root@hpc-master ~]#
```

```
# Check NIS maps are accessible
xdsh compute1,compute2,login,storage "ypcat passwd"
```

```
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "ypcat passwd"
compute1: admin:$6$baTtuxRQSFktHnHf$MwNk0lRPEoizdQKHzkAepEZlujR/sckI6BWXxNMrihVlC47xiAgsACWXm4z5/5rxAMT7Kwk6opN8C1d1rjUU30:1000:1000:Admi
n:/home/admin:/bin/bash
storage: admin:$6$baTtuxRQSFktHnHf$MwNk0lRPEoizdQKHzkAepEZlujR/sckI6BWXxNMrihVlC47xiAgsACWXm4z5/5rxAMT7Kwk6opN8C1d1rjUU30:1000:1000:Admin
:/home/admin:/bin/bash
login: admin:$6$baTtuxRQSFktHnHf$MwNk0lRPEoizdQKHzkAepEZlujR/sckI6BWXxNMrihVlC47xiAgsACWXm4z5/5rxAMT7Kwk6opN8C1d1rjUU30:1000:1000:Admin:/
home/admin:/bin/bash
compute2: admin:$6$baTtuxRQSFktHnHf$MwNk0lRPEoizdQKHzkAepEZlujR/sckI6BWXxNMrihVlC47xiAgsACWXm4z5/5rxAMT7Kwk6opN8C1d1rjUU30:1000:1000:Admi
n:/home/admin:/bin/bash
[root@hpc-master ~]#
```

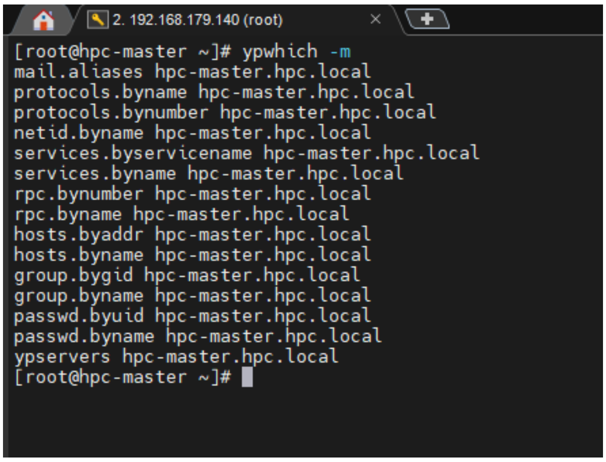
```
# Check nsswitch.conf on all clients
xdsh compute1,compute2,login,storage "grep -E '^passwd|^shadow|^group'
/etc/nsswitch.conf"
```



```
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "grep -E '^passwd|^shadow|^group' /etc/nsswitch.conf"
compute1: passwd:     files nis
compute1: shadow:     files nis
compute1: group:      files nis
storage: passwd:     files nis
storage: shadow:     files nis
storage: group:      files nis
login: passwd:     files nis
login: shadow:     files nis
login: group:      files nis
compute2: passwd:     files nis
compute2: shadow:     files nis
compute2: group:      files nis
[root@hpc-master ~]#
```

4.2 Verify NIS Maps on Server

```
# List all NIS maps and their serving server
ypwhich -m
```

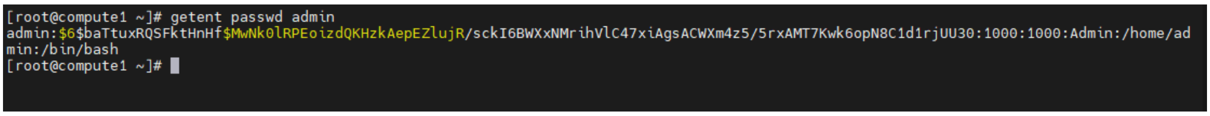


4.3 Verify OS is Using NIS for User Lookups

```
# Run on any client node — most important test
getent passwd admin
```

Expected output:

```
admin:x:1000:1000:Admin:/home/admin:/bin/bash
```



`getent` confirms the OS is actually resolving users through NIS — not just that NIS is reachable. This is the definitive verification test.

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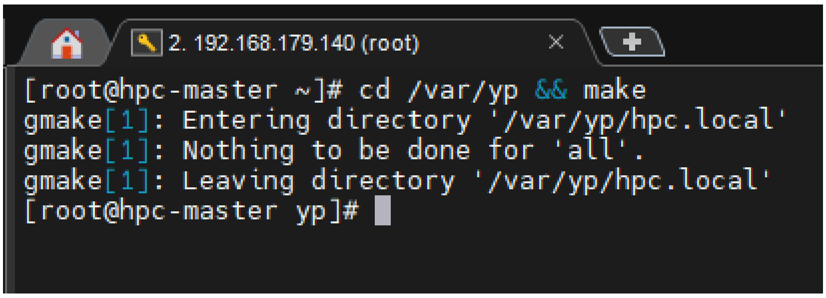
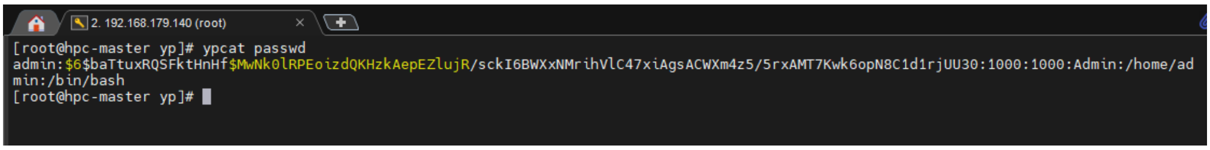
5. Updating NIS Maps

Whenever users are added, modified, or deleted on hpc-master, the NIS maps must be rebuilt and pushed to all clients:

```
# Run on hpc-master after any user/group changes
cd /var/yp && make
```




```
# Verify updated maps
ypcat passwd
```



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7. Key Commands

Purpose	Command
Check NIS domain	`nisdomainname`
Check which server node is bound to	`ypwhich`
List all NIS maps	`ypwhich -m`
View NIS passwd map	`ypcat passwd`
View NIS group map	`ypcat group`
View NIS hosts map	`ypcat hosts`
Test OS NIS user lookup	`getent passwd <username>`
Rebuild NIS maps after changes	`cd /var/yp && make`
Check ypbind status	`systemctl status ypbind`
Check ypserv status	`systemctl status ypserv`
Restart NIS server	`systemctl restart ypserv`
Restart NIS client	`systemctl restart ypbind`